

Market and Patent Research Report.

ProdDev SS25 - Group A

April 14, 2025

1 Introduction

Invasive alien plant species such as *Heracleum mantegazzianum*, *Lupinus polyphyllus*, and *Jacobaea vulgaris* pose significant environmental and public health challenges in Europe. These plants negatively affect native biodiversity, outcompete local flora, and may pose toxicological risks to humans and animals. Conventional removal methods rely on herbicides, which are increasingly restricted by EU environmental legislation (e.g., Regulation (EU) No. 1143/2014), or labor-intensive mechanical clearing. Recent advances in agricultural robotics provide an opportunity to address this issue using autonomous systems that employ machine learning, vision technologies, and non-chemical removal techniques, including electric pulse applications.

2 Market Overview and Technological Landscape

2.1 RootWave (UK) – Electric Weed Control System

- **Technology:** Applies high-frequency electric pulses through contact probes to thermally destroy plant cells from root to tip.
- **Target Species:** Effective on deep-rooted perennials including *Heracleum mantegazzianum*.
- **Advantages:**
 - Non-chemical and environmentally safe.
 - Minimal soil disturbance.
 - Suitable for robotic integration. [Carbon Robotics](#)
- **Use Cases:** Municipalities, ecological reserves, organic agriculture.



Figure 1: Source: RootWave

Video Demonstration:

- [RootWave Demonstration 1](#)
- [RootWave Demonstration 2](#)

2.2 Carbon Robotics – LaserWeeder

- **Technology:** AI-powered machine vision with mounted CO₂ lasers for precision weed eradication.
- **Target Species:** More suitable for row crops. Not designed specifically for tall or toxic plants like hogweed.



Figure 2: Source: RootWave

- **Video Demonstration:** [LaserWeeder Overview](#)

2.3 Odd. Bot – Weed Pulling Robot (NL)

- **Technology:** Selectively detects and pulls weeds using robotic grippers, based on image classification algorithms.
- **Target Species:** Applicable for broadleaf weed species like *Jacobaea vulgaris*.



Figure 3: Source: RootWave

2.4 Drone and Aerial Detection Platforms (DK, DE, UK)

- **Developer:** University of Connecticut (BirdHabitatBot)
- **Technology:** Drones combined with NDVI, hyperspectral imaging, or machine learning can map infestations of hogweed in open fields and forests.
- **Advantages:** Ideal for pre-mapping and targeting robotic ground intervention.



Figure 4: Source: RootWave

2.5 FarmWise – Titan FT-35 (USA)

- **Manufacturer:** [FarmWise](#)
- **Technology:** A self-driving robotic cultivator with vision and mechanical blade system for precision weeding. Optional fertilizer or pesticide application.
- **Target Species:** Broadleaf weeds in vegetable crops.
- **Use Cases:** Commercial vegetable farms.

2.6 Naïo Technologies – Dino (France)

- **Manufacturer:** [Naïo Technologies](#)
- **Technology:** GPS navigation, laser scanners, vision-based selective mechanical weeding.
- **Target Species:** Weeds between and near crop rows.
- **Use Cases:** Organic farms requiring non-chemical weeding.

2.7 ecoRobotix – AVO (Switzerland)

- **Manufacturer:** [ecoRobotix](#)
- **Technology:** Solar-powered robot with precision herbicide spray using machine learning.
- **Target Species:** Isolated weeds among broadleaf crops.
- **Use Cases:** Sustainable farming.

2.8 FarmDroid – FD20 (Denmark)

- **Manufacturer:** [FarmDroid](#)
- **Technology:** GPS-based seeding and blind mechanical weeding.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** Organic grain and vegetable farms.

2.9 Small Robot Company – “Tom & Dick” (UK)

- **Manufacturer:** [Small Robot Company](#)
- **Technology:** Tom scouts and maps, Dick applies electric shock to kill weeds.
- **Target Species:** Broadleaf weeds in high-value crops.
- **Use Cases:** Precision agriculture.

2.10 AutoWeed – Lantana Targeting Robot (Australia)

- **Developer:** [James Cook University](#)
- **Technology:** AI and cameras detect and spray herbicide on *Lantana camara*.
- **Target Species:** *Lantana camara*.
- **Use Cases:** Biodiversity protection.

2.11 BirdHabitatBot – Forest Invasives Robot (USA)

- **Developer:** [University of Connecticut](#)
- **Technology:** Drone-GPS mapping + ground robot to remove invasive shrubs.
- **Target Species:** *Japanese barberry*, *multiflora rose*.
- **Use Cases:** Forest conservation.

2.12 Bosch Deepfield – BoniRob (Germany)

- **Developer:** [Bosch Research](#)
- **Technology:** Vision-guided pneumatic “stamping” system.
- **Target Species:** Early-stage weed seedlings.
- **Use Cases:** High-speed non-chemical cultivation.

2.13 Hydrilla Hunter – Robotic Boat (USA)

- **Developer:** [Northeastern University & CAES](#)
- **Technology:** Hyperspectral aquatic detection and robotic navigation.
- **Target Species:** *Hydrilla*, *Eurasian watermilfoil*.
- **Use Cases:** Freshwater habitat protection.

3 Patent Research Overview

3.1 EP3804483A1 – Tool for Eradication of Invasive Hogweed Species

- **Assignee:** Søren Lauritzen (Denmark)
- **Description:** Manual stem-puncturing tool to induce necrosis in *Heracleum mantegazzianum*.
- **Significance:** Basis for robotic puncturing adaptation.
- **Link:** [EP3804483A1](#)

3.2 WO2021055485A1 – Autonomous Laser Weed Eradication System

- **Assignee:** Carbon Robotics
- **Description:** Robotic platform guiding laser source for weed removal.
- **Limitation:** Not optimized for toxic plants or non-row fields.
- **Link:** [WO2021055485A1](#)

3.3 US20170238460A1 – Weeding Robot and Method

- **Assignee:** Blue River Technology
- **Description:** Selective weeding robot using sensors and mechanical tools.
- **Potential Use:** Adaptable for invasive species.
- **Link:** [US20170238460A1](#)

3.4 EP3744173A1 – Weed Inactivation Device Using Electric Pulses

- **Assignee:** Zasso GmbH
- **Description:** Electrode-based high-voltage weed inactivation.
- **Relevance:** Integrable into autonomous platforms.
- **Link:** [EP3744173A1](#)

3.5 DE102015209879A1 – Device for Damaging Weeds

- **Assignee:** Bosch GmbH
- **Description:** Tool with classification and positioning unit for targeted weed damage.
- **Significance:** Precise, rapid chemical-free weed control.
- **Link:** [DE102015209879A1](#)

3.6 DE4039797A1 – Weed Control Device

- **Assignee:** Robert Bosch GmbH
- **Description:** Continuous weed destruction system interrupted only when crop detected.
- **Significance:** Efficient for automated systems.
- **Link:** [DE4039797A1](#)

Traide Fair Australia

During a recent trade fair visit (14.04.2025) to XAG Sydney Australia, supported by further online research, the **XAG P150 agricultural drone** was identified as a promising solution for the removal of invasive plant species. The drone features a payload capacity of up to 70 kg and uses the **RevoSpray 4 system** to apply herbicides either across wide areas or with pinpoint accuracy. The spraying system delivers fine droplets via adjustable centrifugal nozzles, enabling effective coverage on both sides of plant leaves.

The process is managed using **PIX4Dfields**, a mapping and control software. First, a drone equipped with a high-resolution camera captures detailed images of the area. In the software, the user manually marks invasive species, protected plants, and obstacles. These inputs train the software to recognize similar vegetation patterns using machine learning. The generated prescription map is then uploaded to the spraying drone, which treats only the marked zones with high precision.

Compared to ground-based mowers like the **ROBO MOWER T-90** by NetEvoTech, spraying drones offer several advantages. Mechanical mowing can lead to seed dispersion and is limited in difficult or uneven terrain. Experts advised against the use of mowers in this context and recommended drone application to avoid further spread and ensure better reach. The system is also compatible with drone charging stations from **Enovance**, which can quickly recharge batteries for continuous operation. The total cost for a two-drone system including software is approximately **AUD 50,000**.

Sources:

XAG P150 Product Page: <https://www.xa.com/en/p150>

PIX4Dfields Software Overview: <https://www.pix4d.com/product/pix4dfields>

RevoSpray 4 Spraying System: <https://www.xa.com/en/revospray>

ROBO MOWER T-90 Specs (NetEvoTech): <https://www.netevotech.com/robo-mower-t90>

Enovance Drone Charging: <https://enovance.com.au/>

NetEvoTech

ROBO MOWER T-PREMIUM

Stronger Build with Eco Power Option



Battery Version Available



Choose the battery-powered option of T-PREMIUM for lower noise levels and eco-friendly, emission-free performance.



Reinforced Tracks



Upgraded Dual Blades



Powerful Lift System



Debris Protection

Optional Accessories



Snow Plough



Rotary Brush



Industrial Remote Control



Remote Control with Display and Camera



Tow Hitch



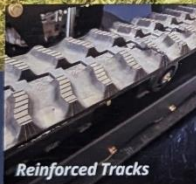
Upgraded Enclosed Design

The advanced enclosed design offers superior protection, enhancing the safety and longevity of both the machine and the operator.

NetEvoTech

ROBO MOWER T-90

Unleash the Power, Conquer Any Terrain



Reinforced Tracks



Upgraded Blades



Powerful Lift System



Debris Protection

Optional Accessories



Cargo Platform



Atomizer 200L



Horizontal and
Vertical Sprinkler 200L



Snow Plough



Rotary Brush



Industrial
Remote Control



Remote Control with
Display and Camera



Tow Hitch

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🌐 www.xag-au.com

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O-Power 21 (Battery Pack)

Model	O-Power 21 (Battery Pack)		
Cell technology	LiFePO ₄	Nominal voltage	76.8V
Dimension (D*W*H)	820*580*295mm	Recommended crate	0.5C
Package size (D*W*H)	1300*730*540mm	USB interface voltage	5V
Weight(Net weight)	174.5kg	USB interface power	65W
Nominal capacity	280Ah	Display	4.3 inches
Nominal energy	21.5kWh	Heat dissipation method	Air cooling
Battery group mode	1P24S	Ingress protection	IP20
Rated charge/discharge current	140A	Working altitude	≤3000m
Max.Charge/Discharge current	200A	Operating temperature	0-50°C
Operating voltage	67.2V - 84.24V	Cycle life	≥3000 Cycles

IBC6000W (AC-DC Charger)

Model	IBC6000W (AC-DC Charger)		
Dimension (D*W*H)	450*223.4*66.2mm	Power	6kW
Weight (Net weight)	8.5 kg	Input method	L+N+PE
Rated input voltage	220Vac	Heat dissipation method	Air cooling
Input voltage range	90Vac-264Vac (90 - 185Vac derating 2000W output)	Ingress protection rating	IP20
Output voltage range	60Vdc - 87.6Vdc	Working altitude	≤3000m
Output current	0 - 70A	Operating temperature	0-50°C

IBC10000W (DC-DC Charger)

Model	IBC10000W (DC-DC Charger)		
Dimension (D*W*H)	550*230*94.5mm	Output power	10kW
Weight (Net weight)	14.5kg	Heat dissipation method	Air cooling
Input voltage range	65Vdc-100Vdc	Ingress protection	IP20
Output voltage range	0.5Vdc-58.8Vdc	Working altitude	≤3000m
Output current	0 - 180A (Default 150A)	Operating temperature	0-50°C





1:29

VoLTE 65%



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PIX4Dfields

Hybrid drone and satellite
mapping software for aerial crop
analysis and precision
agriculture

Saving time, money, and agricultural inputs has
never been easier.

Try for free

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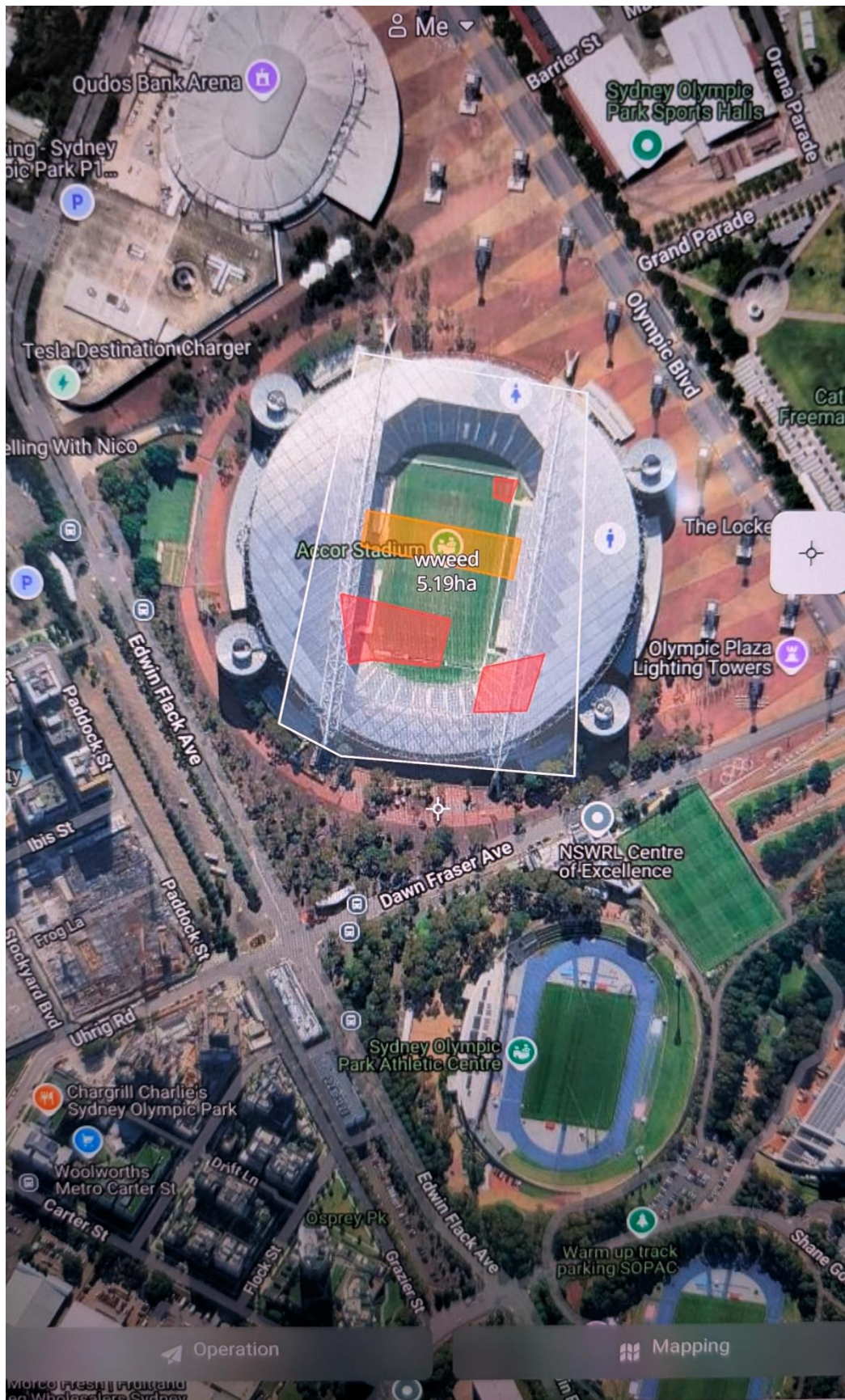


Software available in English, German,
Japanese, Spanish, Chinese, Portuguese,
French, Italian, Czech, Hungarian and Russian.



Available for Windows & macOS





This is how the App looks... Red is no go area and orange is spray area

Requirements List

Course: Product Development – From Need to Market

Institution: University of applied sciences Kaiserslautern

Semester: SS 2025

Intorduction

The following specification outlines the functional and non-functional requirements for the design and development of the robotic equipment described herein. It forms the basis for design decisions, validation, and verification activities.

This document is subject to controlled updates throughout the lifecycle of the project. Any modification must not compromise equipment safety, operational effectiveness, or engineering soundness.

If a requirement does not explicitly list a new date of change, it is to be understood that the date of change is the same as the requirement listed immediately above it.

Abbreviations

HMI	Human-Machine Interface
ROI	Return on Investment
GS	Geprüfte Sicherheit
CE	Conformité Européenne

Legend

F	Fixed requirement
G	Goal requirement
D	Desired requirement

Position	Date of Change	Type of requirement	Requirements List “Autonomous Robot for Invasive Plant Removal”	Created 08.04.2025
			Goal / main function: To autonomously identify, locate, and remove invasive plants in natural environments without harming native vegetation.	Responsible
1	12.04.25		Area of Application	
1.1		F	Suitable for meadows, pastures, and light forest terrain.	
1.2		G	Capable of operating in areas with vegetation cover including bushes and sparse trees.	
1.3		G	Able to overcome common terrain obstacles (stones, bumps, ditches up to 25 cm depth/height).	
1.4		F	Capable of driving through grass and light bushes without performance loss.	
2	12.04.25		Target Plant Types	
2.1		F	Recognizes invasive species such as: perennial lupine, giant hogweed, common ragwort.	
2.2		F	Operates with an AI-based plant recognition system (visual detection with plant database).	
2.3		G	Identification accuracy >= 95%	
2.4	14.04.25	G	Target plants: Height 30–500 cm, stem diameter up to 5 cm.	
3			Removal Method / Effectiveness	
3.1	12.04.25	F	Effectively removes target plant to prevent regrowth (e.g., root removal or neutralization).	
3.2		G	Regrowth prevention effectiveness >= 90% per treated plant.	
3.3		F	Does not damage nearby plants or surrounding soil (non-target impact < 5%).	
4			Mobility and Terrain Compatibility	
4.1	12.04.25	F	Low soil compaction: max ground pressure <= 30 kPa.	
4.2		G	Usable on slightly damp soils without traction loss.	
4.3		G	Operational slope tolerance: up to 20° incline.	
5			Power Supply / Operation Time	

5.1	12.04.25	F	Autonomous operation without grid connection.	
5.2		G	Continuous working time ≥ 6 hours per charge.	
5.3		G	Recharge time ≤ 2 hours.	
5.4		F	Electrical operation: 230V, 50Hz supply for charging.	
5.5	14.04.25	G	Connected electrical load ≤ 1.5 kW.	
6			Safety	
6.1	12.04.25	F	Emergency stop feature accessible and clearly marked.	
6.2		F	Sensors to detect human presence and stop operation automatically.	
6.3		F	Safety perimeter monitoring for unauthorized access.	
6.4		F	Visual alerts when in operation.	
6.5	14.04.25	F	Must detect persons within a radius of at least 5 meters.	
6.6		F	Audible alerts must activate when the working zone is penetrated.	
6.7		F	GS-Certification.	
6.8		F	CE labeling compliant.	
7			Environmental Impact	
7.1		F	Operational temperature range: -10°C to $+40^{\circ}\text{C}$.	
7.2	12.04.25	G	Relative humidity: 10–90% non-condensing.	
7.3		G	Rain and dust tolerant (IP65 level minimum).	
7.4	14.04.25	D	Operation in light snowfall and moderate wind (<40 km/h).	
8			Usability and Handling	
8.1		F	Remote control or app interface for setup and monitoring.	
8.2	12.04.25	G	Interface language selection (See 15)	
8.3		D	Autonomy status feedback and progress visualization.	
9			Transport and Deployment	
9.1		F	Modular and compact form for transport.	
9.2	14.04.25	G	Deployment time < 15 minutes.	
9.3		F	Single person transport with assistance tool. Unit weight ≤ 80 kg.	
10			Cost Targets	

10.1		F	Maximum production cost: €15,000.	
10.2		G	Maintenance cost per year < €800.	
10.3	12.04.25	D	ROI time <= 5 years compared to conventional methods (cost of traditional/manual invasive plant removal methods).	
11			Service Life and Maintenance	
11.1	12.04.25	F	Minimum operational life: 5 years.	
11.2		G	Maintenance interval: every 200 hours of operation.	
11.3		D	Tool-less maintenance access.	
11.4		F	Cleaning duration: manually < 10 minutes.	
12			Recycling and Sustainability	
12.1	14.04.25	G	80% of materials recyclable.	
12.2	12.04.25	D	Eco-friendly packaging and logistics.	
13			Dimensions and Weight	
13.1	14.04.25	F	Max dimensions: 1.2 m (L) × 0.8 m (W) × 1.2 m (H).	
13.2		G	Max transport weight: 50 kg.	
14			Processing Speed	
14.1		F	Average removal time per plant <= 10 seconds.	
14.2		G	Minimum 6 plants removed per minute in dense vegetation.	
14.3		D	Feedback system to verify successful removal per plant.	
15			Languages	
15.1		F	Technical documentation in English.	
15.2		F	Source code documentation in English.	
15.3		F	Instruction manuals in English.	
15.4		G	Instruction manuals available in German, French, Spanish, Russian.	
15.5		F	HMI (screen/app) interface in English.	
15.6		G	HMI interface also available in German, French, Spanish, Russian.	
16			Project Timeline	
16.1		F	Project plan includes timeline, milestones, and documentation phases.	

16.2	F	Timeline to identify critical paths.	
16.3	F	Milestones must include commodity sourcing for testing with specified quantities.	
17		Documentation	
17.1	F	Functional specification including architecture, software, and hardware design with I/O list.	
17.2	F	Alarms test, critical and safety functions report.	
17.3	F	Safety component listing, hazard analysis, and design review.	
17.4	F	Equipment specification with configurable settings and parameters.	
17.5	F	Installation and re-assembly procedure.	
17.6	F	Controls documentation: switches, sensors, indicator functions.	
17.7	F	Mechanical adjustment and format changeover instructions.	
17.8	F	Cleaning and preventative maintenance instructions.	
17.9	F	Troubleshooting procedures.	
17.10	F	Bill of materials.	
17.11	F	Assembly drawings.	
17.12	F	Electrical schematics.	
17.13	F	Software copies of all programs	
18		Network	
18.1	F	All controllers with communication ports must include nameplate with IP, baud rate, serial config, etc.	
18.2	F	Wireless communication must comply with regulatory standards.	
19		Warranty	
19.1	F	Minimum 1-year warranty covering all components.	